Product Requirement Document & Test Suite

Feature Module: User Authentication & Payment Gateway

Module Priority: P0 (Critical)

1. Requirements Overview

• REQ-01: System must allow secure user login via OAuth2. (Priority: P0)
• REQ-02: Multi-factor authentication (MFA) must be enforced for admin roles. (Priority: P0)
• REQ-03: Payment processing via Stripe must support tokenized credit cards. (Priority: P0)
• REQ-04: System shall log all failed login attempts for security auditing. (Priority: P1)

2. Test Cases & Traceability Matrix

• TC-101 (Maps to REQ-01): Verify successful login with valid credentials. (Type: Positive)
• TC-102 (Maps to REQ-01): Verify login failure with invalid password. (Type: Negative)
• TC-103 (Maps to REQ-02): Verify MFA challenge prompt on admin login. (Type: Positive)
• TC-104 (Maps to REQ-03): Verify successful payment execution with Stripe. (Type: Positive)
• TC-105 (Maps to REQ-04): Verify security audit logs record lockouts correctly. (Type: Negative)

3. Summary Statistics

Total Requirements: 4
Total Test Cases: 5
Mapped Requirements: 4
Critical (P0) Requirements: 3
Unmapped Critical Requirements: 0
Negative Test Cases: 2